

YEAR 7 MATHEMATICS • CHAPTER 7

Decimals

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|--|--|
| 7.01 Ordering decimals | 7.02 Decimals and fractions |
| 7.03 Adding and subtracting decimals | 7.04 Multiplying and dividing by powers of 10 |
| 7.05 Using a known product | 7.06 Multiplying decimals |
| 7.07 Dividing decimals by whole numbers | 7.08 Dividing by a decimal |
| 7.09 Fractions to decimals | 7.10 Rounding decimals |
| 7.11 Decimals and money | |

Name _____

Class _____

Teacher _____

How to use this booklet

- Each lesson starts with **learning intentions** and **success criteria**. Write the **key terms** as your teacher explains them, and use the **notes** box.
- Work through the levels in order: Level 1, then Level 2, then Level 3. Extension questions stretch you further.
- Some questions ask you to **draw** a diagram, graph or construction. Use the space given, a ruler and a sharp pencil.
- At the end of each lesson, **show off your skill**: choose one question, then tear off the ticket and hand it in.
- Missed a lesson? Each lesson ends with a **summary** that has a worked example for every level. Start there, then try the questions.

WE DO

Grey boxes with a dark border. Work with your teacher. Copy what they write on the board.

YOU DO

Blue boxes. Work on your own. Show your working in the box and write your answer.

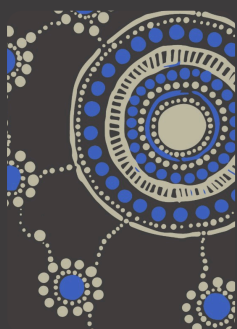
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In this chapter you will

- order decimals
- convert between decimals and fractions
- add, subtract, multiply and divide decimals
- multiply and divide by powers of 10
- round decimals and write recurring decimals
- solve money problems

NSW Mathematics K–10 Syllabus (2022), Stage 4 Number and algebra: MA4-FRC-C-01 represents and operates with fractions, decimals and percentages to solve problems.



Acknowledgement of artwork

The artwork used throughout this booklet series was created by **Marni Tuala**. We thank Marni for sharing this artwork with our school community.

Acknowledgement of Country

We acknowledge the Bundjalung people, the Traditional Custodians of the land on which Kingscliff High School stands, and pay our respects to Elders past and present.

LESSON
7.01

Ordering decimals

Date _____

Learning intentions

We are learning to:

- compare and order decimals

Success criteria

I can:

- ☐ use place value to compare decimals
- ☐ fill with zeros so decimals have the same number of places
- ☐ arrange decimals in ascending and descending order

Key terms Write the definition as your teacher explains it.

Ascending

Descending

Notes

Level 1 · Getting started

WE DO

With your teacher: copy what they write on the board.

Example 1

Which is larger: 0.7 or 0.65?

Example 2

Which is larger: 3.998 or 34.89?

YOU DO

On your own: Set A. Which is larger?

1 0.7 or 0.65

2 2.5 or 2.45

3 0.09 or 0.1

4 14.3 or 14.29

5 5.06 or 5.6

6 0.333 or 0.34

YOU DO On your own: Set B. Write the place value of the 7.

1 3.7

4 70.2

2 0.07

5 6.0071

3 12.457

6 17.5

Level 2 · Building up

WE DO With your teacher: copy what they write on the board.

Example 3 Write the place value of 8 in 12.084.

Example 4 Arrange 34.98, 36.86, 3.998, 38.141, 340, 34.89 in ascending order.

YOU DO On your own: Set C. Arrange in ascending order.

1 34.98, 36.86, 3.998, 38.141, 340, 34.89

3 2.1, 2.01, 2.11, 2.101, 2.011

2 0.5, 0.45, 0.405, 0.54, 0.045

4 7.6, 7.06, 7.066, 7.606, 7.66



Level 3 · Challenge

Finished Set D? Try the extension on the next page.

WE DO With your teacher: copy what they write on the board.**Example 5** Arrange 1.015, 1.293, 1.1015, 1.239, 1.006 in descending order.
Example 6 Write a decimal between 2.35 and 2.36.
YOU DO On your own: Set D. Order and compare.**1** Arrange 1.015, 1.293, 1.1015, 1.239, 1.006 in descending order.
3 Four runners: 12.08 s, 12.8 s, 12.018 s, 12.108 s. Who won?
2 Write a decimal between 2.35 and 2.36.
4 Which is larger: 0.1 or 0.099 999? Explain.

✕ Show off your skill: tear along this line and hand it to your teacher ✕

YOU DO Show off your skillChoose **ONE** question. Show your working.

Name _____ Class _____

Level 1

Which is larger: 0.4 or 0.38?

Level 2

Arrange 5.2, 5.02, 5.22 in ascending order.

Level 3

Arrange 0.3, 0.303, 0.033, 0.33 in descending order.

Extension · Stretch yourself

Worked example

Write the decimal halfway between 3.47 and 3.5.

- $3.5 = 3.50$
- Halfway between 3.47 and 3.50 is 3.485

Answer: 3.485

YOU DO

On your own: read the worked example, then try these.

- E1** Find the decimal halfway between 0.6 and 0.61.

- E2** How many decimals with 2 decimal places are between 4.3 and 4.4?

Lesson summary

Missed the lesson? Start here.

- Remember**
1. Line up the decimal points.
 2. Fill empty places with zeros.
 3. Compare from the left, place by place.
 4. Ascending: smallest first. Descending: largest first.

Level 1

0.7 or 0.65

1. 0.70 vs 0.65

Answer: 0.7

Level 2

Place of 8 in 12.084

1. Hundredths

Answer: 8 hundredths

Level 3

Ascending: 34.98, 3.998, 340...

1. Compare whole numbers first

Answer: 3.998, 34.89, 34.98, 36.86, 38.141, 340

Extension

Halfway 3.47 and 3.5

1. 3.470 and 3.500

Answer: 3.485

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LESSON
7.02

Decimals and fractions

Date _____

Learning intentions

We are learning to:

- convert between decimals and fractions

Success criteria

I can:

- ☐ write a decimal as a fraction over 10, 100 or 1000
- ☐ simplify the fraction
- ☐ write a fraction with denominator 10, 100 or 1000 as a decimal

Key terms Write the definition as your teacher explains it.

Tenths

Hundredths

Thousandths

Notes

Level 1 · Getting started

WE DO

With your teacher: copy what they write on the board.

Example 1

Write 0.5 as a simplified fraction.

Example 2

Write 0.89 as a fraction.

YOU DO

On your own: Set A. Write each decimal as a fraction.

1 0.3

4 0.7

2 0.89

5 0.21

3 0.09

6 0.003

YOU DO On your own: Set B. Write each fraction as a decimal.

1 $\frac{4}{10}$

4 $\frac{11}{1000}$

2 $\frac{13}{100}$

5 $\frac{9}{10}$

3 $\frac{7}{100}$

6 $\frac{125}{1000}$

Level 2 · Building up**WE DO** With your teacher: copy what they write on the board.**Example 3** Write 0.444 as a simplified fraction.**Example 4** Write $\frac{4}{10}$ as a decimal.**YOU DO** On your own: Set C. Write as a simplified fraction or mixed numeral.

1 0.5

3 0.444

5 2.6

2 0.25

4 0.08

6 3.125



Level 3 · Challenge

Finished Set D? Try the extension on the next page.

WE DO With your teacher: copy what they write on the board.**Example 5** Write $\frac{7}{100}$ as a decimal.**Example 6** Write $\frac{11}{1000}$ as a decimal.
YOU DO On your own: Set D. Convert.**1** Write $\frac{3}{5}$ as a decimal (make the denominator 10).
3 Write $\frac{9}{250}$ as a decimal (make it 1000).
2 Write $\frac{7}{20}$ as a decimal (make it 100).
4 Which is larger: 0.35 or $\frac{3}{8}$? Explain.

✕ Show off your skill: tear along this line and hand it to your teacher ✕

YOU DO Show off your skillChoose **ONE** question. Show your working.

Name _____ Class _____

Level 1

Write 0.7 as a fraction.

Level 2Write $\frac{23}{100}$ as a decimal.
Level 3

Write 0.35 as a simplified fraction.

Extension · Stretch yourself

Worked example

Write $\frac{17}{25}$ as a decimal.

1. $\times 4$ top and bottom: $\frac{68}{100}$

Answer: 0.68

YOU DO

On your own: read the worked example, then try these.

E1 Write $\frac{3}{40}$ as a decimal.

E2 Write 4.075 as a simplified mixed numeral.

Lesson summary

Missed the lesson? Start here.

Remember

1. Count the decimal places: 1 → tenths, 2 → hundredths, 3 → thousandths.
3. Simplify.

2. Write the digits over 10, 100 or 1000.

4. To go back: make the denominator 10, 100 or 1000.

Level 1

0.5

1. $\frac{5}{10}$

Answer: $\frac{1}{2}$

Level 2

0.444

1. $\frac{444}{1000}$

Answer: $\frac{111}{250}$

Level 3

$\frac{11}{1000}$

1. 3 decimal places

Answer: 0.011

Extension

$\frac{17}{25}$

1. $\frac{68}{100}$

Answer: 0.68

This space is left blank on purpose. It is the back of your exit ticket.

LESSON
7.03

Adding and subtracting decimals

Date _____

Learning intentions

We are learning to:

- add and subtract decimals

Success criteria

I can:

- ☐ line up the decimal points
- ☐ fill empty places with zeros
- ☐ estimate to check the answer

Key terms Write the definition as your teacher explains it.**Decimal point****Estimate**

Notes

Level 1 · Getting started**WE DO**

With your teacher: copy what they write on the board.

Example 1 $12.35 + 4.53$ **Example 2** $9.23 - 6.851$ **YOU DO**

On your own: Set A. Evaluate.

1 $3.4 + 2.5$ **2** $12.35 + 4.53$ **3** $0.7 + 0.8$ **4** $6.9 - 2.4$ **5** $10.5 - 3.2$ **6** $5 + 0.75$

YOU DO On your own: Set B. Evaluate.

1 $9.23 - 6.851$

4 $12.1 - 0.07$

2 $4.6 + 0.58$

5 $0.568 + 23$

3 $8 - 3.45$

6 $20 - 0.02$

Level 2 · Building up**WE DO** With your teacher: copy what they write on the board.

Example 3 $0.568 + 23 + 4.027$

Example 4 $214.33 - 109.84$

YOU DO On your own: Set C. Evaluate. Set out in columns.

1 $214.33 - 109.84$

3 $0.568 + 23 + 4.027 - 16.28$

5 $300 - 47.38$

2 $1600.8 - 562.9$

4 $45.07 + 3.9 + 128.625$

6 $7.2 - 3.456 + 1.8$



Level 3 · Challenge

Finished Set D? Try the extension on the next page.

WE DO With your teacher: copy what they write on the board.**Example 5** $1453.6 + 1287.31 - 2344.4$ **Example 6** $12.35 + 4.53 + 0.56 + 3.125 + 24.7 + 20.09$ **YOU DO** On your own: Set D. Evaluate.**1** $12.35 + 4.53 + 0.56 + 3.125 + 24.7 + 20.09$ **3** Kim had \$50. She spent \$12.85 and \$7.40. How much is left?**2** $1453.6 + 1287.31 - 2344.4$ **4** Two parcels weigh 3.75 kg and 2.9 kg. How much heavier is the first?

✕ Show off your skill: tear along this line and hand it to your teacher ✕

YOU DO

Show off your skill

Choose **ONE** question. Show your working.

Name _____ Class _____

Level 1

$2.7 + 1.6$

Level 2

$15.3 - 8.75$

Level 3

$6.05 + 13.9 - 0.87$

Extension · Stretch yourself

Worked example

Find the missing number: $4.8 - ? + 2.35 = 5$

1. $4.8 + 2.35 = 7.15$
2. $7.15 - ? = 5$

Answer: 2.15

YOU DO

On your own: read the worked example, then try these.

- E1** Find the missing number: $? + 3.07 - 1.5 = 10$.

- E2** Using each of the digits 1 to 6 once, make two decimals of the form $\square.\square\square$ with the largest possible sum.

Lesson summary

Missed the lesson? Start here.

- Remember**
1. Line up the decimal points.
 2. Fill gaps with zeros.
 3. Add or subtract as for whole numbers.
 4. Put the point in line in the answer.

Level 1

$$12.35 + 4.53$$

1. Line up points

Answer: 16.88

Level 2

$$9.23 - 6.851$$

1. $9.230 - 6.851$

Answer: 2.379

Level 3

$$214.33 - 109.84$$

1. Borrow carefully

Answer: 104.49

Extension

$$4.8 - ? + 2.35 = 5$$

1. $7.15 - 5$

Answer: 2.15

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LESSON
7.04

Multiplying and dividing by powers of 10

Date _____

Learning intentions

We are learning to:

- multiply and divide decimals by 10, 100 and 1000

Success criteria

I can:

- ☐ move digits one place for each zero
- ☐ $\times$ moves digits left (bigger); $\div$ moves digits right (smaller)
- ☐ add zeros as place holders

Key terms Write the definition as your teacher explains it.**Power of 10****Place holder**

Notes

Level 1 · Getting started

WE DO

With your teacher: copy what they write on the board.

Example 1

7.54×10

Example 2

7.54×100

YOU DO

On your own: Set A. Evaluate.

1 7.54×10

2 7.54×100

3 0.3×10

4 4.82×1000

5 0.056×100

6 12.5×10

YOU DO On your own: Set B. Evaluate.

1 $13.9 \div 10$

4 $45.2 \div 1000$

2 $13.9 \div 100$

5 $0.8 \div 100$

3 $6 \div 10$

6 $350 \div 1000$

Level 2 · Building up**WE DO** With your teacher: copy what they write on the board.**Example 3** 7.54×1000 **Example 4** $13.9 \div 10$ **YOU DO** On your own: Set C. Evaluate.

1 $13.9 \div 1000$

3 $2.5 \times 10\,000$

5 $0.4 \times 100 \div 10$

2 0.0072×1000

4 $9.1 \div 10\,000$

6 $85 \div 100 \times 1000$



Level 3 · Challenge

Finished Set D? Try the extension on the next page.

WE DO With your teacher: copy what they write on the board.**Example 5** $13.9 \div 100$ **Example 6** $13.9 \div 1000$ **YOU DO** On your own: Set D. Problems.**1** A pen costs \$1.35. Find the cost of 100 pens.**3** 1000 sheets of paper stack 96 mm high. How thick is one sheet?**2** Ten friends share a \$87.50 bill. How much each?**4** Find the missing number: $3.47 \times ? = 3470$

✕ Show off your skill: tear along this line and hand it to your teacher ✕

YOU DO

Show off your skill

Choose **ONE** question. Show your working.

Name _____ Class _____

Level 1

3.8×10

Level 2

$52.6 \div 100$

Level 3

0.049×1000

 Extension · Stretch yourself
 Worked example

Evaluate 0.025×400 without a calculator.

1. $400 = 4 \times 100$
2. $0.025 \times 100 = 2.5$
3. 2.5×4

Answer: 10

YOU DO

On your own: read the worked example, then try these.

- E1** Evaluate 3.6×2000 .

- E2** Evaluate $48 \div 300$ (hint: $\div 3$ then $\div 100$).

Lesson summary

Missed the lesson? Start here.

Remember

1. Count the zeros in 10, 100, 1000.
3. $\div$: move the digits that many places right.

2. $\times$: move the digits that many places left.
4. Use zeros to fill empty places.

 Level 1

7.54×100

1. 2 places left

Answer: 754

 Level 2

$13.9 \div 100$

1. 2 places right

Answer: 0.139

 Level 3

0.0072×1000

1. 3 places left

Answer: 7.2

 Extension

0.025×400

1. 2.5×4

Answer: 10

This space is left blank on purpose. It is the back of your exit ticket.

LESSON
7.05

Using a known product

Date _____

Learning intentions

We are learning to:

- use a known product to multiply decimals

Success criteria

I can:

- ☐ count the decimal places in the question
- ☐ put the same number of decimal places in the answer
- ☐ estimate to check the size of the answer

Key terms Write the definition as your teacher explains it.**Product****Decimal places**

Notes

Level 1 · Getting started**WE DO** With your teacher: copy what they write on the board.**Example 1** Given $42 \times 76 = 3192$, find 4.2×76 .**Example 2** Find 4.2×7.6 .**YOU DO** On your own: Set A. Given $42 \times 76 = 3192$, find:

1 4.2×76

2 42×7.6

3 4.2×7.6

4 420×76

5 0.42×76

6 42×0.76

YOU DOOn your own: Set B. Given $15 \times 23 = 345$, find:

1 1.5×23

4 15×0.23

2 1.5×2.3

5 150×2.3

3 0.15×23

6 0.15×0.23

**Level 2 · Building up****WE DO**

With your teacher: copy what they write on the board.

Example 3Find 0.42×760 .**Example 4**Find 4.2×0.76 .**YOU DO**On your own: Set C. Given $42 \times 76 = 3192$, find:

1 0.42×760

3 0.42×0.76

5 420×0.076

2 4.2×0.76

4 0.042×76

6 $3192 \div 76$



Level 3 · Challenge

Finished Set D? Try the extension on the next page.

WE DO With your teacher: copy what they write on the board.**Example 5** Find 42×7.6 .
Example 6 Find 0.42×0.76 .
YOU DO On your own: Set D. Given $125 \times 36 = 4500$, find:**1** 1.25×36

3 0.125×0.36

2 12.5×3.6

4 $45 \div 3.6$

✕ Show off your skill: tear along this line and hand it to your teacher ✕

YOU DO Show off your skillChoose **ONE** question. Show your working.

Name _____ Class _____

Level 1Given $23 \times 14 = 322$, find 2.3×14 .
Level 2Find 2.3×1.4 .
Level 3Find 0.23×0.14 .

 Extension · Stretch yourself
 Worked example

Given $17 \times 24 = 408$, find $408 \div 1.7$.

- $408 \div 17 = 24$
- Dividing by a number 10 times smaller gives an answer 10 times bigger

Answer: 240

YOU DO

On your own: read the worked example, then try these.

- E1** Given $17 \times 24 = 408$, find $40.8 \div 2.4$.

- E2** Given $17 \times 24 = 408$, find $0.408 \div 0.017$.

Lesson summary

Missed the lesson? Start here.

- Remember**
- Ignore the decimal points and use the known product.
 - Count the decimal places in the question.
 - The answer has the same total.
 - Estimate to check.

 Level 1

$$4.2 \times 76$$

- 1 decimal place

Answer: 319.2

 Level 2

$$4.2 \times 7.6$$

- 2 decimal places

Answer: 31.92

 Level 3

$$0.42 \times 0.76$$

- 4 decimal places

Answer: 0.3192

 Extension

$$408 \div 1.7$$

- 10×24

Answer: 240

This space is left blank on purpose. It is the back of your exit ticket.

LESSON
7.06

Multiplying decimals

Date _____

Learning intentions

We are learning to:

- multiply decimals

Success criteria

I can:

- ☐ multiply as whole numbers first
- ☐ count the total decimal places in the question
- ☐ place the decimal point in the answer

Key terms Write the definition as your teacher explains it.**Product****Decimal places**

Notes

Level 1 · Getting started

WE DO With your teacher: copy what they write on the board.**Example 1** 2.75×6 **Example 2** 0.5×1.2 **YOU DO** On your own: Set A. Evaluate.

1 2.75×6

2 0.92×5

3 3×9.7

4 72.23×4

5 0.6×3

6 1.5×8

YOU DO On your own: Set B. Evaluate.

1 0.5×1.2

4 2.5×0.2

2 0.3×0.4

5 0.05×0.02

3 0.7×0.9

6 1.1×1.1

Level 2 · Building up**WE DO** With your teacher: copy what they write on the board.

Example 3 6.1×1.2

Example 4 3.25×0.41

YOU DO On your own: Set C. Evaluate.

1 6.1×1.2

3 4.67×1.1

5 0.84×2.5

2 3.25×0.41

4 12.4×0.35

6 9.05×0.06



Level 3 · Challenge

Finished Set D? Try the extension on the next page.

WE DO With your teacher: copy what they write on the board.**Example 5** 0.05×0.02 **Example 6** 4.67×1.1 **YOU DO** On your own: Set D. Problems.**1** Find the area of a rectangle 4.5 m by 3.2 m.**3** Find 0.2^3 .**2** Petrol costs \$1.89 per litre. Find the cost of 42.5 L (to the nearest cent).**4** Without calculating, is 0.9×0.8 more or less than 0.8? Explain.

✕ Show off your skill: tear along this line and hand it to your teacher ✕

YOU DO Show off your skillChoose **ONE** question. Show your working.

Name _____ Class _____

Level 1

4.3×2

Level 2

0.6×0.7

Level 3

2.35×1.4

Extension · Stretch yourself

Worked example

Evaluate $1.2 \times 1.2 \times 1.2$.

- $1.2 \times 1.2 = 1.44$
- $1.44 \times 1.2 = 1.728$

Answer: 1.728

YOU DO

On your own: read the worked example, then try these.

- E1** Evaluate $0.3 \times 0.3 \times 0.3 \times 0.3$.

- E2** Find two decimals whose product is 0.24.

Lesson summary

Missed the lesson? Start here.

- Remember**
1. Multiply the numbers ignoring the points.
 2. Count all decimal places in the question.
 3. Give the answer that many decimal places.
 4. Estimate: $6.1 \times 1.2 \approx 6 \times 1 = 6$.

Level 1

$$2.75 \times 6$$

- $275 \times 6 = 1650$
- 2 decimal places

Answer: 16.5

Level 2

$$0.05 \times 0.02$$

- $5 \times 2 = 10$
- 4 decimal places

Answer: 0.001

Level 3

$$3.25 \times 0.41$$

- $325 \times 41 = 13\,325$
- 4 places

Answer: 1.3325

Extension

$$1.2^3$$

- 1.44×1.2

Answer: 1.728

This space is left blank on purpose. It is the back of your exit ticket.

LESSON
7.07

Dividing decimals by whole numbers

Date _____

Learning intentions

We are learning to:

- divide a decimal by a whole number

Success criteria

I can:

- ☐ use short division, keeping the decimal point in line
- ☐ add zeros to keep dividing
- ☐ solve sharing problems

Key terms Write the definition as your teacher explains it.**Quotient****Short division**

Notes

Level 1 · Getting started

WE DO With your teacher: copy what they write on the board.**Example 1** $762.4 \div 2$ **Example 2** $97.6 \div 8$ **YOU DO** On your own: Set A. Evaluate.

1 $762.4 \div 2$

2 $8.4 \div 4$

3 $9.3 \div 3$

4 $1.25 \div 5$

5 $0.63 \div 7$

6 $12.6 \div 6$

YOU DO On your own: Set B. Evaluate.

1 $97.6 \div 8$

4 $0.72 \div 9$

2 $195.6 \div 3$

5 $31.2 \div 4$

3 $4.55 \div 5$

6 $6.3 \div 7$

Level 2 · Building up**WE DO** With your teacher: copy what they write on the board.

Example 3 $195.6 \div 3$

Example 4 $2.75 \div 4$

YOU DO On your own: Set C. Evaluate. Add zeros if needed.

1 $2.75 \div 4$

3 $7 \div 4$

5 $12.1 \div 5$

2 $3.5 \div 8$

4 $0.9 \div 6$

6 $1.8 \div 12$



Level 3 · Challenge

Finished Set D? Try the extension on the next page.

WE DO With your teacher: copy what they write on the board.**Example 5** $3.5 \div 8$ **Example 6** Share \$50.40 among 6 people.**YOU DO** On your own: Set D. Problems.**1** Share \$50.40 among 6 people.**3** 8 cans of drink cost \$11.60. Find the cost of one can.**2** A 4.5 m plank is cut into 4 equal pieces. How long is each?**4** A runner covers 21.1 km in 5 equal laps. How long is each lap?

✕ Show off your skill: tear along this line and hand it to your teacher ✕

YOU DO Show off your skillChoose **ONE** question. Show your working.

Name _____ Class _____

Level 1

$6.8 \div 2$

Level 2

$14.4 \div 6$

Level 3

$5.3 \div 4$

 Extension · Stretch yourself
 Worked example

The mean of 3 decimals is 2.45. Two of them are 1.8 and 3.1. Find the third.

1. Total = $3 \times 2.45 = 7.35$

2. $7.35 - 1.8 - 3.1$

Answer: 2.45

YOU DO

On your own: read the worked example, then try these.

- E1** The mean of 4 numbers is 5.6. Three are 4.2, 7.1 and 5.5. Find the fourth.

- E2** Find $10 \div 8$ and $10 \div 16$. What do you notice?

Lesson summary

Missed the lesson? Start here.

Remember

1. Set out short division.
2. Put the decimal point in the answer above the point in the question.
3. Divide each digit, carrying remainders.
4. Add zeros after the point to keep going.

 Level 1

$762.4 \div 2$

1. Divide each digit

Answer: 381.2

 Level 2

$97.6 \div 8$

1. $9 \div 8 = 1$ r1, $17 \div 8 = 2$ r1, $16 \div 8 = 2$

Answer: 12.2

 Level 3

$2.75 \div 4$

1. $2.750 \div 4$

Answer: 0.6875

 Extension

Mean 2.45, find third

1. $7.35 - 4.9$

Answer: 2.45

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LESSON
7.08

Dividing by a decimal

Date _____

Learning intentions

We are learning to:

- divide by a decimal

Success criteria

I can:

- ☐ multiply both numbers by 10, 100 or 1000 to make the divisor whole
- ☐ then divide by the whole number
- ☐ check that the answer makes sense

Key terms Write the definition as your teacher explains it.**Divisor****Equivalent division**

Notes

Level 1 · Getting started

WE DO With your teacher: copy what they write on the board.**Example 1** $12.5 \div 0.5$ **Example 2** $12.72 \div 0.4$ **YOU DO** On your own: Set A. Evaluate.

1 $12.5 \div 0.5$

2 $6 \div 0.2$

3 $3.6 \div 0.4$

4 $0.8 \div 0.1$

5 $4.2 \div 0.7$

6 $10 \div 0.5$

YOU DO On your own: Set B. Evaluate.

1 $12.72 \div 0.4$

4 $0.48 \div 0.6$

2 $2.45 \div 0.5$

5 $1.5 \div 0.05$

3 $9.6 \div 0.3$

6 $0.9 \div 0.09$

Level 2 · Building up**WE DO** With your teacher: copy what they write on the board.

Example 3 $6.9 \div 0.03$

Example 4 $0.508 \div 0.02$

YOU DO On your own: Set C. Evaluate.

1 $6.9 \div 0.03$

3 $4 \div 0.8$

5 $2.4 \div 0.004$

2 $0.508 \div 0.02$

4 $7.35 \div 0.15$

6 $0.126 \div 0.3$



Level 3 · Challenge

Finished Set D? Try the extension on the next page.

WE DO With your teacher: copy what they write on the board.**Example 5** $4 \div 0.8$ **Example 6** How many 0.25 L cups can be filled from 3.5 L?
YOU DO On your own: Set D. Problems.**1** How many 0.25 L cups can be filled from 3.5 L?
3 Ribbon costs \$0.45 per metre. How many metres for \$6.30?
2 A car travels 157.5 km on 12.6 L of fuel. How many km per litre?
4 Explain why $5 \div 0.5$ is larger than 5.

✕ Show off your skill: tear along this line and hand it to your teacher ✕

YOU DO

Show off your skill

Choose **ONE** question. Show your working.

Name _____ Class _____

Level 1

$8 \div 0.4$

Level 2

$5.4 \div 0.6$

Level 3

$0.36 \div 0.09$

 Extension · Stretch yourself
 Worked example
Evaluate $0.0144 \div 0.012$.1. $\times 1000$: $14.4 \div 12$

Answer: 1.2

YOU DO

On your own: read the worked example, then try these.

E1 Evaluate $3.24 \div 0.018$.**E2** Is $8 \div 0.1$ bigger or smaller than 8×10 ? Explain.**Lesson summary**

Missed the lesson? Start here.

Remember 1. Make the divisor a whole number.

3. Then divide as usual.

2. Multiply both numbers by the same power of 10.

4. Dividing by a number less than 1 makes the answer bigger.

 Level 1 $12.5 \div 0.5$ 1. $125 \div 5$

Answer: 25

 Level 2 $12.72 \div 0.4$ 1. $127.2 \div 4$

Answer: 31.8

 Level 3 $6.9 \div 0.03$ 1. $690 \div 3$

Answer: 230

 Extension $0.0144 \div 0.012$ 1. $14.4 \div 12$

Answer: 1.2

This space is left blank on purpose. It is the back of your exit ticket.

LESSON
7.09

Fractions to decimals

Date _____

Learning intentions

We are learning to:

- convert fractions to decimals, including recurring decimals

Success criteria

I can:

- ☐ divide the numerator by the denominator
- ☐ recognise a **recurring** decimal
- ☐ write recurring decimals using dot notation

Key terms Write the definition as your teacher explains it.**Terminating decimal****Recurring decimal**

Notes

Level 1 · Getting started

WE DO

With your teacher: copy what they write on the board.

Example 1

 $\frac{4}{5}$ as a decimal

Example 2

 $\frac{3}{8}$ as a decimal

YOU DO

On your own: Set A. Write as a decimal.

1 $\frac{4}{5}$

4 $\frac{1}{2}$

2 $\frac{1}{4}$

5 $\frac{2}{5}$

3 $\frac{3}{4}$

6 $\frac{7}{20}$

YOU DO

On your own: Set B. Write as a decimal using short division.

1 $\frac{3}{8}$

4 $\frac{9}{40}$

2 $\frac{5}{8}$

5 $\frac{7}{8}$

3 $\frac{1}{16}$

6 $\frac{3}{25}$

**Level 2 · Building up****WE DO**

With your teacher: copy what they write on the board.

Example 3 $\frac{2}{3}$ as a decimal**Example 4** $\frac{5}{9}$ as a decimal**YOU DO**

On your own: Set C. Write as a recurring decimal using dot notation.

1 $\frac{2}{3}$

3 $\frac{1}{3}$

5 $\frac{1}{6}$

2 $\frac{5}{9}$

4 $\frac{5}{11}$

6 $\frac{4}{7}$



Level 3 · Challenge

Finished Set D? Try the extension on the next page.

WE DO With your teacher: copy what they write on the board.**Example 5** $\frac{5}{11}$ as a decimal**Example 6** $\frac{1}{6}$ as a decimal
YOU DO On your own: Set D. Think it through.**1** Order $\frac{2}{3}$, 0.6, 0.67, $\frac{5}{8}$ from smallest to largest.
3 Write $\frac{1}{7}$ as a decimal. How many digits repeat?
2 Write $\frac{1}{9}$, $\frac{2}{9}$ and $\frac{3}{9}$ as decimals. Predict $\frac{7}{9}$.
4 Which fractions with denominators 2 to 10 give terminating decimals?

✕ Show off your skill: tear along this line and hand it to your teacher ✕

YOU DO Show off your skillChoose **ONE** question. Show your working.

Name _____ Class _____

Level 1Write $\frac{3}{5}$ as a decimal.
Level 2Write $\frac{7}{8}$ as a decimal.
Level 3Write $\frac{1}{3}$ as a recurring decimal.

 Extension · Stretch yourself

 Worked example

Write $0.\dot{4}$ as a fraction.

1. $0.444... = 4 \times 0.111...$

2. $0.111... = \frac{1}{9}$

Answer: $\frac{4}{9}$

YOU DO

On your own: read the worked example, then try these.

E1 Write $0.\dot{7}$ as a fraction.

E2 Write $0.\dot{1}\dot{8}$ ($0.181\ 818...$) as a fraction. Hint: $0.\dot{0}\dot{1} = \frac{1}{99}$.

Lesson summary

Missed the lesson? Start here.

Remember

1. Divide the top by the bottom.
3. Terminating: the division ends.

2. Add zeros after the decimal point to keep going.
4. Recurring: digits repeat forever. Show them with a dot: $0.\dot{6}$.

 Level 1

$$\frac{4}{5}$$

1. $4 \div 5$

Answer: 0.8

 Level 2

$$\frac{3}{8}$$

1. $3.000 \div 8$

Answer: 0.375

 Level 3

$$\frac{2}{3}$$

1. $2 \div 3 = 0.666...$

Answer: $0.\dot{6}$

 Extension

$$0.\dot{4}$$

1. 4 ninths

Answer: $\frac{4}{9}$

This space is left blank on purpose. It is the back of your exit ticket.

LESSON
7.10

Rounding decimals

Date _____

Learning intentions

We are learning to:

- round decimals to a given number of decimal places

Success criteria

I can:

- ☐ find the last digit to keep
- ☐ look at the next digit: 5 or more rounds up
- ☐ round to the nearest tenth, hundredth or thousandth

Key terms Write the definition as your teacher explains it.

Decimal places

Nearest tenth

Notes

Level 1 · Getting started

WE DO With your teacher: copy what they write on the board.**Example 1** Round 406.28 to the nearest tenth.**Example 2** Round 125.724 to 2 decimal places.**YOU DO** On your own: Set A. Round to the nearest tenth (1 decimal place).

1 406.28

2 3.14

3 12.35

4 0.96

5 7.05

6 2345.876

YOU DO On your own: Set B. Round to 2 decimal places.

1 125.724

4 19.999

2 3.8967

5 4.1049

3 0.555

6 78 654.056

Level 2 · Building up**WE DO** With your teacher: copy what they write on the board.**Example 3** Round 3.8967 to the nearest hundredth.**Example 4** Round 2345.876 to 1 decimal place.**YOU DO** On your own: Set C. Round as shown.

1 678.4309 (3 d.p.)

3 0.0456 (2 d.p.)

5 0.3333 (1 d.p.)

2 5.6 (whole number)

4 12.9951 (2 d.p.)

6 9.95 (1 d.p.)



Level 3 · Challenge

Finished Set D? Try the extension on the next page.

WE DO With your teacher: copy what they write on the board.**Example 5** Round 78 654.056 to 2 decimal places.
Example 6 Round 678.4309 to the nearest thousandth.
YOU DO On your own: Set D. Rounding in context.**1** Write $2 \div 3$ to 2 decimal places.
3 A number rounds to 4.7 (1 d.p.). What are the smallest and largest it could be (2 d.p.)?
2 A bill of \$47.23 is shared by 3 people. How much each, to the nearest cent?
4 Round $\pi = 3.141\ 59\dots$ to 1, 2 and 3 decimal places.

✕ Show off your skill: tear along this line and hand it to your teacher ✕

YOU DO Show off your skillChoose **ONE** question. Show your working.

Name _____ Class _____

Level 1

Round 5.67 to 1 decimal place.

Level 2

Round 12.345 to 2 decimal places.

Level 3

Round 0.8996 to the nearest thousandth.

Extension · Stretch yourself

Worked example

Round 0.049 95 to 3 significant figures.

- Significant figures start at the first non-zero digit: 4
- Keep 4, 9, 9; next digit is 5, round up

Answer: 0.0500

YOU DO

On your own: read the worked example, then try these.

- E1** Round 23 461 to 2 significant figures.

- E2** Round 0.006 074 to 2 significant figures.

Lesson summary

Missed the lesson? Start here.

- Remember**
- Find the last digit you keep.
 - 5 or more: round up. 4 or less: leave.

- Look at the next digit to the right.
- Drop the digits after.

Level 1

406.28 to 1 d.p.

- Next digit 8

Answer: 406.3

Level 2

3.8967 to 2 d.p.

- Next digit 6: 3.89 → 3.90

Answer: 3.90

Level 3

678.4309 to 3 d.p.

- Next digit 9

Answer: 678.431

Extension

0.049 95 to 3 s.f.

- Round up

Answer: 0.0500

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LESSON
7.11

Decimals and money

Date _____

Learning intentions

We are learning to:

- solve money problems with decimals

Success criteria

I can:

- ☐ choose the operations the problem needs
- ☐ set out working clearly with \$ and 2 decimal places
- ☐ round money to the nearest cent or 5 cents

Key terms Write the definition as your teacher explains it.

Total

Change

Notes

Level 1 · Getting started

WE DO

With your teacher: copy what they write on the board.

Example 1

A club orders 16 pizzas at \$19.70 each and 10 at \$15.50 each. Find the total cost.

Example 2

Riley bought 825 bricks for \$1105.50. How much did one brick cost?

YOU DO

On your own: Set A. Evaluate.

1 $\$4.50 + \3.75

2 $\$20 - \6.85

3 $5 \times \$2.40$

4 $\$18.60 \div 3$

5 $\$100 - \37.45

6 $\$1.95 \times 4$

YOU DO On your own: Set B. Round to the nearest 5 cents.

1 \$17.83

4 \$25.06

2 \$4.12

5 \$0.52

3 \$9.98

6 \$13.37

Level 2 · Building up**WE DO** With your teacher: copy what they write on the board.

Example 3 Mia saved \$220. She spent \$18.80, \$116.75, \$25 and \$37.70. How much is left?

Example 4 Mia wants a \$59.95 backpack. How much must she borrow?

YOU DO On your own: Set C. Solve.

1 16 pizzas at \$19.70 and 10 at \$15.50. Find the total.

3 Find the change from \$50 for \$12.95, \$8.40 and \$19.99.

5 3 friends share a \$127.35 meal equally.

2 825 bricks cost \$1105.50. Find the cost of one brick.

4 Apples are \$4.80/kg. Find the cost of 2.5 kg.

6 A \$65 game is on sale for 20% off. Find the sale price.



Level 3 · Challenge

Finished Set D? Try the extension on the next page.

WE DO With your teacher: copy what they write on the board.**Example 5** Round \$17.83 to the nearest 5 cents.
Example 6 Find the change from \$50 for items costing \$12.95, \$8.40 and \$19.99.
YOU DO On your own: Set D. Multi-step problems.

- 1 Mia saved \$220. She spent \$18.80 on fares, \$116.75 on a ticket, \$25 on a program and \$37.70 on food. How much must she borrow to buy a \$59.95 backpack?

- 2 A phone plan costs \$35 a month plus \$0.15 per MB over the limit. Last month Sam went 64 MB over. Find the bill.

- 3 Which is the better buy: 1.5 L for \$3.60 or 2 L for \$4.60?

- 4 Tickets cost \$12.50 for adults and \$7.80 for children. Find the cost for 2 adults and 3 children.

✕ Show off your skill: tear along this line and hand it to your teacher ✕

YOU DO

Show off your skill

Choose **ONE** question. Show your working.

Name _____ Class _____

Level 1Find $4 \times \$3.65$.
Level 2

Find the change from \$20 for \$7.45 and \$6.90.

Level 3

Six tickets cost \$76.50. Find the cost of one ticket.

Extension · Stretch yourself

Worked example

A shop sells pens at 3 for \$4.20 or 5 for \$6.50. Which is cheaper per pen?

1. $4.20 \div 3 = \$1.40$

2. $6.50 \div 5 = \$1.30$

Answer: 5 for \$6.50

YOU DO

On your own: read the worked example, then try these.

E1 You need at least 7 pens. Find the cheapest way to buy them using the packs.

E2 A pack of 12 costs \$14.40. Is it better than 5 for \$6.50?

Lesson summary

Missed the lesson? Start here.

Remember 1. Read carefully and list what you know.
3. Write money with \$ and 2 decimal places.

2. Choose +, −, × or ÷ for each step.
4. Check the answer is sensible.

Level 1

Pizza order

1. $16 \times 19.70 = 315.20$

2. $10 \times 15.50 = 155.00$

Answer: \$470.20

Level 2

One brick

1. $1105.50 \div 825$

Answer: \$1.34

Level 3

Mia borrows

1. $220 - 198.25 = 21.75$

2. $59.95 - 21.75$

Answer: \$38.20

Extension

Pens

1. \$1.40 vs \$1.30

Answer: 5 for \$6.50

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